

Muhammad Zaid

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Summary

Final-year Computer Science student at UET Lahore with experience in Data Science, Machine Learning, and Python backend development. Built production-scale ETL pipelines processing 400K+ records, developed an Urdu-English translation model achieving BLEU-48, and created an offline AI assistant using local LLMs and RAG architecture. Seeking an entry-level Data Analyst, ML Engineer, or Python Developer role.

Projects

NextGen AI Desktop Assistant (Final Year Project)

Python, PySide6, Ollama, ChromaDB, Speech Recognition

- Built a fully offline AI desktop assistant using local LLMs via Ollama, enabling private, internet-free AI interactions.
- Implemented contextual memory with ChromaDB (RAG), allowing the assistant to recall conversation history across sessions.
- Designed a modular PySide6 GUI with structured voice-command workflows and a command-execution testing pipeline.
- Designed architecture to support real-time voice interaction and system-level automation (file management, app launching).

Urdu-to-English Machine Translation using Pakistani Drama Subtitles

Python, Hugging Face, PyTorch

- Curated and preprocessed a 160K+ sentence corpus from 500 Pakistani drama episodes.
- Used facebook/m2m100.600M for English-to-Urdu data augmentation, then fine-tuned facebook/m2m100.418M for Urdu-to-English translation.
- **Achieved a BLEU score of 48** on domain-specific conversational data, significantly outperforming the zero-shot baseline (~18 BLEU).

CineMatch Movie Recommender System

Python, Streamlit, Pandas, Scikit-Learn

- Built a content-based recommender on the TMDB dataset (10,000+ movies) by engineering a unified feature vector from title, genres, keywords, overview, cast, crew, and tagline.
- Applied TF-IDF vectorization and cosine similarity to retrieve the top-5 most relevant recommendations in under 200 ms.
- Integrated the TMDB API for live poster fetching and deployed the app on Streamlit Cloud for public access.

Experience

Associate Software Engineer, Big Bytes Insight Tech

Jan 2026 – Mar 2026

- Engineered 2 production Scrapy spiders that collected **400K+ vehicle listings and 400K+ property records**, eliminating manual data collection entirely.
- Designed a two-stage MySQL schema (staging + cleaned tables) reducing data load errors by enforcing validation before promotion to the clean layer.
- Built end-to-end ETL pipelines using Scrapy, Pandas, and MySQL, reducing manual processing time by 80%.

Education

University of Engineering and Technology (UET), Lahore

Expected Graduation: May 2026

Bachelor of Computer Science

CGPA: 3.20/4.00

Relevant Coursework: Artificial Intelligence, Machine Learning, Natural Language Processing, Introduction to Data Science

Skills

Programming Language: Python, C++ ,SQL

Databases: PostgreSQL, MySQL

Data Science & ML: Pandas, NumPy, Scikit-Learn, PyTorch, NLTK, Hugging Face Transformers

Web & APIs: FastAPI, Scrapy, Selenium, BeautifulSoup

Visualization: Matplotlib, Plotly, Seaborn, Tableau

Tools: Git, Docker, Jupyter Notebook, VS Code

Certifications

2025: IBM Data Analyst Certificate – Coursera

2026: FASTAPI – CampusX

References

Available upon request.